of the instrument by means of the wires. ☞ See that the zinc does not touch either of the carbons in the cell by tilting to one side.☜

The immersed surface of the zinc plate may be increased or diminished by changing the position of its clamp up or down, thus increasing or diminishing at pleasure the QUANTITY of electricity evolved. Before the lower half of the zinc is entirely eaten away, it may be turned in the clamp and the other half immersed, thus utilizing the whole plate.

When removing the zinc from either the rubber-cup or the battery-cell, be careful that no mercury or fluid drops on any part of the apparatus; the acid will stain the wood and eat off the nickel-plating, and, if allowed to dry under or around the hydrostat, will crystallize and prevent its being fluid-tight. Always leave the cell open a few minutes after using, to permit the escape of the hydrogen gas generated during action. This gas must escape in some way, and if the hydrostat be screwed down immediately, the confined gas will force its way up between the carbons and the brass frame to which they are attached, and the fluid will follow by capillary attraction through the air-spaces thus made.

To prevent irregular working of the battery the zinc should be kept well amalgamated,—*i.e.*, well covered with mercury. For this purpose some bisulphate of mercury is added to the solution, which obviates the necessity of keeping quicksilver in the rubber cup. Always replace the zinc in the rubber cup after using the battery.

If the trembler, or interrupter, should not start